MAD-1 Project Report

1. Student Details

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About Me

I am pursuing the BS Degree in Data Science and Applications from IIT Madras. I have a strong interest in web development, artificial intelligence, and data science. This project helped me understand Flask application development, relational databases, authentication, CRUD operations, and responsive web interface design.

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2. Project Details

Project Title

Trekking Management System (TMA)

Problem Statement

The objective of this project is to develop a web-based Trekking Management System that enables administrators to manage trekking events, staff, users, and bookings efficiently. The system also provides users with an easy platform to browse available treks and make bookings while allowing staff members to manage assigned trekking activities.

Approach

The application is developed using the Flask framework following a modular project structure.

The system implements role-based access for three different users:

- Administrator
- Staff
- User

SQLite is used for persistent data storage through SQLAlchemy ORM. Jinja2 templates are used for rendering dynamic HTML pages, while Bootstrap and CSS are used to create a responsive and user-friendly interface.

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3. AI / LLM Declaration

This project was developed with assistance from **ChatGPT (GPT-5)**.

AI assistance was primarily used for:

- Flask project structure guidance
- HTML and CSS layout improvements
- SQLAlchemy model generation
- Debugging application errors
- Template design suggestions
- Documentation preparation
- Code optimization suggestions

The approximate AI/LLM contribution is **25–30%**, mainly for development assistance and documentation.

The application architecture, integration of all modules, testing, debugging, UI customization, and final implementation were completed manually.

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4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend Web Framework
SQLAlchemy	ORM for SQLite Database
SQLite	Database
Jinja2	HTML Template Rendering

HTML5	Frontend Structure
CSS3	Styling
Bootstrap 5	Responsive UI
Flask-Login	User Authentication
Werkzeug Security	Password Hashing
Python	Programming Language

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5. Database Schema

Tables

User

- id
- name
- email
- password
- role

Trek

- id
- trek_name
- location
- duration
- difficulty
- price
- slots
- description

Booking

- id
- user_id
- trek_id
- booking_date

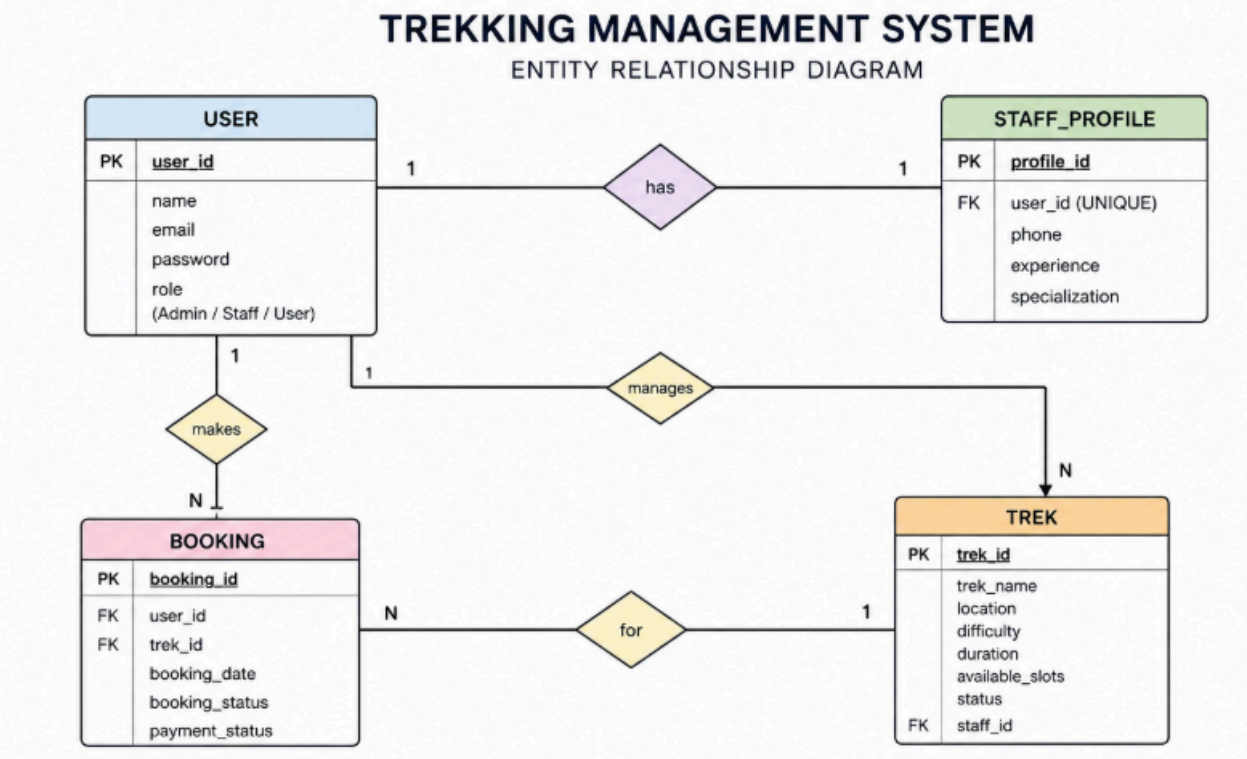
- status

Staff

- id
- name
- email
- password

Relationships

- One User → Many Bookings
- One Trek → Many Bookings
- One Staff → Many Assigned Treks (if applicable)



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6. Application Routes

Route	Method	Description
/	GET	Home Page
/login	GET, POST	User Login
/register	GET, POST	User Registration
/logout	GET	Logout
/admin/dashboard	GET	Admin Dashboard
/admin/treks	GET	Manage Treks
/admin/users	GET	Manage Users
/admin/staff	GET	Manage Staff
/admin/bookings	GET	View Bookings
/staff/dashboard	GET	Staff Dashboard
/staff/treks	GET	Assigned Treks
/staff/participants	GET	Trek Participants
/user/dashboard	GET	User Dashboard
/user/treks	GET	View Treks
/user/bookings	GET	My Bookings

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7. Project Architecture

Folder Structure

```
TMA
│
├── app
│   ├── routes
│   │   admin.py
│   │   auth.py
│   │   staff.py
│   │   user.py
```

```
│   │
│   ├── templates
│   ├── static
│   ├── models.py
│   └── __init__.py
│
├── create_admin.py
├── run.py
├── requirements.txt
└── README.md
```

Features Implemented

Admin

- Secure Login
- Dashboard
- Add Trek
- Edit Trek
- Delete Trek
- Manage Users
- Manage Staff
- View Bookings

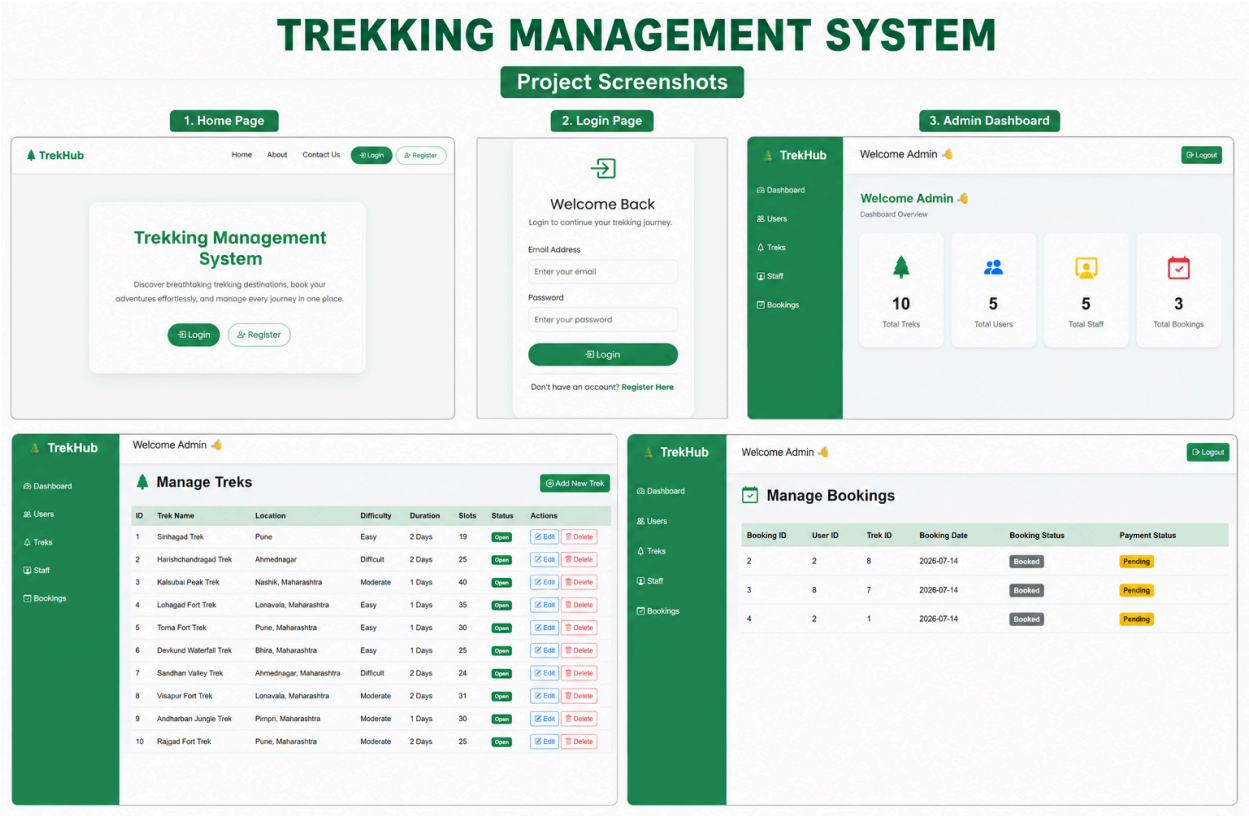
Staff

- Login
- Dashboard
- View Assigned Treks
- View Participants
- Update Trek Details
- Profile

User

- Registration
- Login
- Dashboard
- View Treks
- Book Trek
- View Booking History
- Profile

8. Screenshots



11. GitHub Repository

https://github.com/23f3002563/TMA/tree/master

12. Video Demonstration

Drive Link:

https://drive.google.com/file/d/1MpA8AFuItLvLKoIuHKNIaCSEXUfomLWl/view?usp=sharing

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